EXPERIMENT - 16

16.Implement a Sobel algorithm using Open CV to filter the input image.

Aim

To implement the Sobel algorithm using OpenCV in Python to detect edges in an input image.

Procedure

1. Import the OpenCV library.
2. Read the input image using cv2.imread().
3. Resize the image to fit the screen.
4. Convert the color image into a grayscale image.
5. Apply the Sobel operator to detect horizontal and vertical edges.
6. Combine both edge images to obtain the final output.
7. Display the input image and the Sobel-filtered output image.

Program:

```
import cv2

img = cv2.imread("image.jpg")

if img is None:
   print("Image not found")
else:
   img = cv2.resize(img, (500, 350))
   gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

   sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
   sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

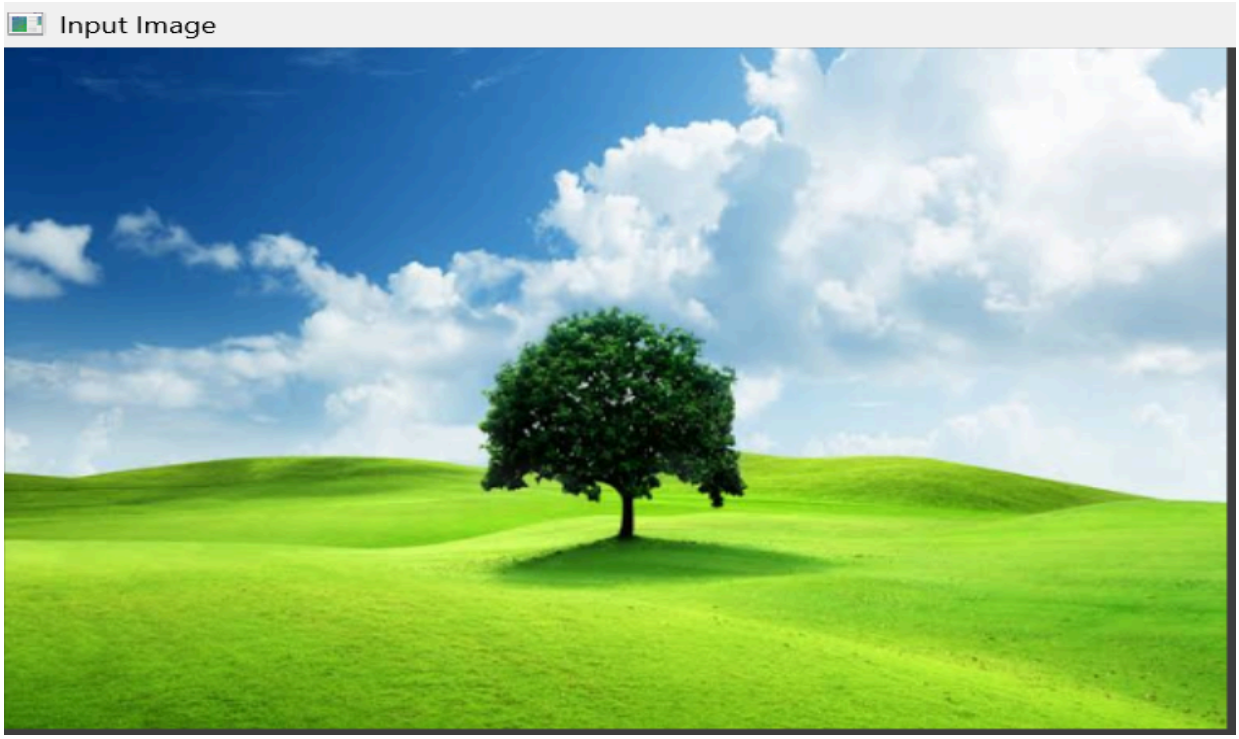
   sobel_x = cv2.convertScaleAbs(sobel_x)
   sobel_y = cv2.convertScaleAbs(sobel_y)

   output = cv2.addWeighted(sobel_x, 0.5, sobel_y, 0.5, 0)

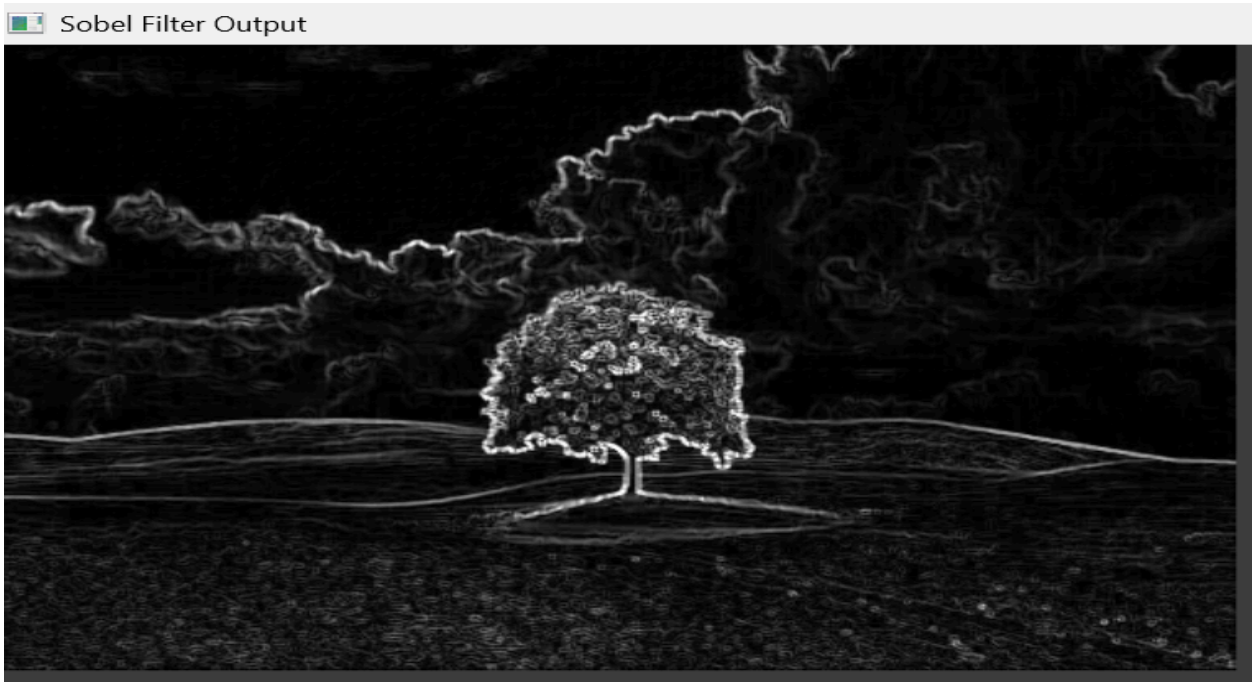
   cv2.imshow("Input Image", img)
   cv2.imshow("Sobel Filter Output", output)

   cv2.waitKey(0)
   cv2.destroyAllWindows()
```

Sample Input:



Sample Output:



Result

The Sobel algorithm was successfully implemented using OpenCV to detect edges in the input image.